Lihua Fu
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Education

Virginia Tech, Blacksburg, VA — Jan 2024 – Expected May 2027
Ph.D. in Environmental Design and Planning
(Specialization: Construction Engineering and Management)
Advisor: Dr. Lu Zhang
GPA: 3.94/4.00
Dissertation: An AI-Powered, Human-Centered Digital Twin for Resilient Infrastructure Planning and Design: Stakeholder Value-Aware Generative Design for Data Centers

Virginia Tech, Blacksburg, VA — Aug 2026 – Expected May 2027
Master of Engineering in Computer Science and Applications

Chongqing University, Chongqing, China — Jun 2023
Master of Engineering in Civil Engineering
Advisor: Dr. Jiepeng Liu
Thesis: Intelligent Modeling and Dimensional Inspection Method of Complex Steel Structure Engineering

Chongqing University, Chongqing, China — Jun 2020
Bachelor of Science in Statistics
Thesis: Research on Automatic Recognition Algorithm of Prefabricated Component Point Cloud Based on Deep Learning

Research Interests

- Digital twins
- Artificial intelligence
- Generative design
- Prompt engineering
- Resilient infrastructure systems
- Human-building interaction
- Multisector stakeholder decision-making
- Multi-agent systems

Publications

Journal Articles

1. **Fu, L.** and Zhang, L. Forthcoming. "GADM-APE: A Genetic Algorithm-Driven, Definition-Integrated, Multistep Automatic Prompt Engineer Framework for Disaster Tweet Classification." *Journal of Computing in Civil Engineering* **(IF: 5.8; Q1 in Engineering, Civil)**. https://doi.org/10.1061/JCCEE5.CPENG-8049.
2. **Fu, L.** and Zhang, L. (2026). "Multi-label disaster tweet classification for human value identification via IDAP-APE: Example prompt construction enabled by inaccuracy-driven active-prompt and automatic prompt engineer." *Progress in Disaster Science* **(IF: 4; Q1 in Geosciences, Multidisciplinary)**. 31, 100616. https://doi.org/10.1016/j.pdisas.2026.100616.
3. **Fu, L.**, Zhang, L., and Sadri, A. M. (2026). "A Prompting Framework for GPT-Based Twitter/X User Classification in the Context of Disasters." *Journal of Computing in Civil Engineering* **(IF: 5.8; Q1 in Engineering, Civil)**. 40(1), 04025129. https://doi.org/10.1061/JCCEE5.CPENG-6751.
4. **Fu, L.**, Xing, Z., Cheng, G., Li, D., Cui, N., and Chen, Y. F. (2022). "Terrestrial laser scanning assisted dimensional quality assessment for space frame components." *Measurement* **(IF: 6.1; Q1 in Engineering, Multidisciplinary)**. 204, 112067. https://doi.org/10.1016/j.measurement.2022.112067.

5. Liu, J., **Fu, L.**, Cheng, G., Li, D., Zhou, J., Cui, N., and Chen, Y. F. (2022). "Automated BIM Reconstruction of Full-Scale Complex Tubular Engineering Structures Using Terrestrial Laser Scanning." *Remote Sensing* **(IF: 4.3; Q1 in Geosciences, Multidisciplinary)**. 14(7), 1659. https://doi.org/10.3390/rs14071659.

6. Ma, X., Zhang, S., Cheng, G., **Fu, L.**, Hu, S., and Li, Y. (2023). "Automated dimensional quality assessment of precast laminated panels based on 3D laser scanning." *Journal of Civil and Environmental Engineering* (Chinese journal). 46(1), 102-109. https://dx.doi.org/10.11835/j.issn.2096-6717.2023.050.

7. **Fu, L.** and Zhang, L. (2026). "Understanding Multisector Stakeholder Value Systems in Data Center Development via LLM-Based Social Media Mining." *Journal of Construction Engineering and Management* **(IF: 5.5; Q1 in Construction & Building Technology)**. Under review.

8. Edusei, E. A., Zhang, L., **Fu, L.**, Schenk, T., and Hall, R. (2026). "Understanding Stakeholder Values in Data Center Development: Alignments, Tensions, and Implications for Collaborative Governance." *Cities* **(IF: 7.0; Q1 in Urban Studies)**. Under review.

9. **Fu, L.**, Zhang, L., and Wang, X. "Multisector Stakeholder Consensus Building in Data Center Development: A Social Media Data (SMD)-Seeded, LLM-Powered Multi-Agent Negotiation Framework." To be submitted.

Peer-Reviewed Conference Proceedings

1. **Fu, L.**, and Zhang, L. (2026). "Mining Twitter/X with Large Language Models to Analyze Multisector Resilience Framings during Hurricane Laura." *23rd International Conference on Smart and Sustainable Built Environment (SASBE 2026)*. In press.

2. **Fu, L.**, Zhang, L., and Wang, X. (2026). "Multisector Stakeholder Consensus Building in Data Center Development: An SNA-Seeded, LLM-Based Multi-Agent Negotiation Framework." *ASCE International Conference on Computing in Civil Engineering (i3CE 2026)*. In press.

3. **Fu, L.** and Zhang, L. (2026). "Multi-Label Disaster Tweet Classification for Human Value Identification via IDAP-APE: Example Prompt Construction Enabled by Inaccuracy-Driven Active-Prompt and Automatic Prompt Engineer." *ASCE International Conference on Computing in Civil Engineering (i3CE 2026)*. In press.

4. **Fu, L.** and Zhang, L. (2026). "GADM-APE: A Genetic Algorithm-Driven, Definition-Integrated, Multistep Automatic Prompt Engineer Framework for Disaster Tweet Classification." *ASCE International Conference on Computing in Civil Engineering (i3CE 2026)*. In press.

5. **Fu, L.**, Zhang, L., and Sadri, A. M. (2025). "An LLM-Based Approach for Stakeholder Classification in Disaster Contexts." *ASCE International Conference on Computing in Civil Engineering (i3CE 2025)*. https://doi.org/10.1061/9780784486436.006.

Research Grants

Virginia Water Resources Research Center (VWRRC) Student Research Grant, USGS **$10,000**
"Evaluating Watershed Hydro-Resilience under Rapid Data Center Growth in Virginia through a Geo-Embedded Explainable Machine Learning Framework"
PRINCIPAL INVESTIGATOR Jan 2027 – Jan 2028
Co-PIs: Dr. Lu Zhang; Dr. Landon Marston

- Led the preparation of the proposal and budget; obtained funding from VWRRC.
- Will develop a weighted geospatial machine learning framework to identify hydro-sensitive watersheds and deliver an interactive decision-support tool to inform data center siting across Virginia.

BioBuild Interdisciplinary Graduate Education Program, Virginia Tech *Under Review*
"Forest-Edge-Inspired Engineering of Data Center Campus Design"
STUDENT INVESTIGATOR Spring 2027
PI: Dr. Lu Zhang; Co-Student Investigator: Shuqi Hu; Funding: GRA support

- Contributed to the preparation of the proposal submission.
- Will develop a bio-inspired framework that translates forest-edge ecology into whole data center campus planning, integrating AI-based generative design and digital-twin simulation to reduce heat, noise, and emissions at community boundaries.

Research Projects

VISTA-DC: An AI-Enabled, Digital-Twin-Based, Human-in-the-Loop Generative Scenario Platform for Data Center Development, Sponsor: M.C. Dean, Inc. USA
RESEARCH ASSISTANT Jul 2026 – Present

- Leading the development of a multi-objective optimization module that identifies the best trade-offs among competing objectives in data center design under county-specific regulations.
- Designed a multi-agent LLM pipeline that automatically generates code-compliant building information models (BIM) from the optimized scenarios, enabling rapid design iteration.
- Developed an immersive human-in-the-loop prototype that enables stakeholders to review and guide AI-powered 3D BIM regeneration.

Integrating Multisector Stakeholder Values for Collaborative Data Center Planning and Policymaking, Sponsor: +Policy Scholar Award, Institute for Society, Culture and Environment (ISCE), Virginia Tech USA
RESEARCH ASSISTANT 2025 – 2026

- Led the development of an LLM-assisted social media analytics pipeline that discovers stakeholder values and quantifies sector-level value priorities in data center development.
- Developed an LLM-based multi-agent negotiation framework that simulates consensus building among stakeholder sectors to support shared decision-making.

A Value-Driven Multi-Sector Stakeholder Decision-Making Framework to Support Disaster Resilient Communities, Sponsor: National Science Foundation (NSF) USA
RESEARCH ASSISTANT Jan 2024 – Dec 2024

- Developed LLM-based prompting frameworks (e.g., IDAP-APE and GADM-APE) to classify social media users into stakeholder groups and identify the human values expressed in their posts.
- Analyzed how value priorities vary across stakeholder sectors and shift across disaster phases.
- Adapted these frameworks to measure stakeholder value systems in data center development.

Proposal Writing Experience

2026 – [Virginia Water Resources Research Center / U.S. Geological Survey (USGS)] "Evaluating Watershed Hydro-Resilience under Rapid Data Center Growth in Virginia through a Geo-Embedded Explainable Machine Learning Framework." **$10,000**. **Role:** PI (*Funded*)

2026 – [M.C. Dean, Inc.] "VISTA-DC: An AI-Enabled, Digital-Twin-Based, Human-in-the-Loop Generative Scenario Platform for Data Center Development." **$100,000**. **Role:** Assisted with Proposal Writing / Research Assistant; **PI:** Dr. Lu Zhang (*Funded*)

2025 – [+Policy Scholar Award, ISCE, Virginia Tech] "Integrating Multisector Stakeholder Values for Collaborative Data Center Planning and Policymaking." **$10,000**. **Role:** Assisted with Proposal Writing / Research Assistant; **PI:** Dr. Lu Zhang (*Funded*)

2025 – [National Science Foundation, Infrastructure Systems and People (ISP)] "From Conflicts to Alignment: A Collaborative Values Integration and Alignment (COVIA) Framework for Human-Centered Data Center Infrastructure Planning and Design." **$750,000**. **Role:** Assisted with Proposal Writing; **PI:** Dr. Lu Zhang (*Under Review*)

2026 – [BioBuild Interdisciplinary Graduate Education Program, Virginia Tech] "Forest-Edge-Inspired Engineering of Data Center Campus Design." **GRA support**. **Role:** Student Investigator; **PI:** Dr. Lu Zhang (*Under Review*)

2026 – [William T. Grant Foundation] "Reconnecting Low-Income Youth to Learning After Disasters: Educational Continuity Practices in Geographically Isolated Appalachian Communities." **Role:** Assisted with Proposal Writing (*Under Review*)

2025 – [National Science Foundation, Strengthening American Infrastructure (SAI)] "SAI: Collaborative Stakeholder Values Integration for Data Center Infrastructure Planning and Design." **$750,000**. **Role:** Assisted with Proposal Writing; **PI:** Dr. Lu Zhang (*Not Funded*)

2024 – [National Science Foundation, Improving Undergraduate STEM Education (IUSE)] "A Value-Driven Course Evaluation Framework to Support Student Engaged Learning." **$400,000**. **Role:** Assisted with Proposal Writing; **PI:** Dr. Lu Zhang (*Not Funded*)

2026 – [4-VA Collaborative Research, Virginia Tech] "AI-Enabled, Stakeholder-Integrated, Context-Sensitive Data Center Planning Framework." **$35,000**. **Role:** Assisted with Proposal Writing; **PI:** Dr. Lu Zhang (*Not Funded*)

2026 – [Global Change Center Seed Grant, Virginia Tech] "Advancing Understanding of Impact of Data Center Infrastructure Growth on Human-Environment Systems." **$20,000**. **Role:** Assisted with Proposal Writing; **PI:** Dr. Lu Zhang (*Not Funded*)

2024 – [Coalition for Smart Construction Seed Grant, Virginia Tech] "Harnessing the Power of Generative Artificial Intelligence to Transform Operations and Workforce Development in the Construction Industry." **$50,000**. **Role:** Assisted with Proposal Writing; **PI:** Dr. Lu Zhang (*Not Funded*)

Invited Talks and Professional Presentations

Invited presentation "LLM-Assisted Social Media Mining: Data Sources, Collection, and LLM-Based Methods." Presented at Virginia Tech Carilion School of Medicine, Roanoke, VA, Aug 2026.

- Presented how social media can serve as a research data source, and introduced data collection methods and self-developed LLM-based data mining methods to graduate students.

Conference presentation "Multisector Stakeholder Consensus Building in Data Center Development: A Social Media Data (SMD)-Seeded, LLM-Powered Multi-Agent Negotiation Framework." Presented at *ASCE International Conference on Computing in Civil Engineering (i3CE 2026)*, Incheon, South Korea, Jun 2026.

- Presented a multi-agent LLM system in which stakeholder agents are modeled from real-world social media discourse and negotiate over a multi-layer network toward consensus on data center development.

Conference presentation "IDAP-APE: An Inaccuracy-Driven Active-Prompt and Automatic Prompt Engineer Framework for Multi-Label Tweet Classification." Presented at *ASCE International Conference on Computing in Civil Engineering (i3CE 2026)*, Incheon, South Korea, Jun 2026.

- Introduced an inaccuracy-driven exemplar selection mechanism that extends automatic prompt engineering to multi-label identification of human values in social media posts.

Conference presentation "GADM-APE: A Genetic Algorithm-Driven, Definition-Integrated, Multistep Automatic Prompt Engineer Framework for Tweet Classification." Presented at *ASCE International Conference on Computing in Civil Engineering (i3CE 2026)*, Incheon, South Korea, Jun 2026.

- Introduced a new automatic prompt engineering approach by optimizing definition-integrated, multistep prompts for different text classification tasks through a genetic algorithm search.

Conference presentation "An LLM-Based Approach for Stakeholder Classification in Disaster Contexts." Presented at *ASCE International Conference on Computing in Civil Engineering (i3CE 2025)*, New Orleans, LA, May 2025.

- Presented a three-step prompting framework that classifies social media users into stakeholder groups using only usernames and profile descriptions.

Teaching Experience

Teaching Assistantships

Myers-Lawson School of Construction, Virginia Tech — Spring 2025 – Present

- CEM 3084 — Construction Economy (Undergraduate Level), Fall 2025, Spring 2026, Fall 2026
  - Graded assignments and in-class quizzes and provided individualized feedback for 120 undergraduate students across two sections.
  - Advised and assessed student final business-plan projects.
  - Guided students on course concepts and assignment requirements.
  - Arranged accommodations to ensure every student had a fair opportunity to demonstrate learning.
- BC 2024/5524 — Construction Principles II (Undergraduate and Graduate Level), Spring 2025
  - Graded course assignments and provided individual written feedback for 42 students.
  - Advised students on course projects and helped coordinate course-related matters.

Guest Lectures

Myers-Lawson School of Construction, Virginia Tech — Nov 2026

- Will present the ongoing resilient data center planning project and demonstrate the AI-Enabled, Digital-Twin-Based, Human-in-the-Loop Generative Scenario Platform to graduate students in Sustainable Infrastructure Systems.

Carilion School of Medicine, Virginia Tech — Aug 2026

- Provided a lecture on "LLM-Assisted Social Media Mining: Data Sources, Collection, and LLM-Based Methods" to introduce social media data sources, data collection methods, and existing and self-developed LLM-based analysis strategies to graduate students.

Efforts on Improving Teaching Effectiveness

Graduate Certificate in Future Professoriate, Virginia Tech — 2025

- Completed graduate training in Contemporary Pedagogy, Preparing the Future Professoriate, and the GTA Training Workshop.

Mentoring Experience

Graduate Research Mentor, Virginia Tech — 2026 – Present

- Mentees: **Emmanuel Edusei**, Ph.D. candidate, Urban and Environmental Design and Planning; **Emily Thompson** and **Lucas Peretti**, Ph.D. students, Virginia Tech Carilion School of Medicine; and **Emily Gaskill**, M.P.H. student, Virginia Tech Carilion School of Medicine.
- Provide three doctoral students and one master's student with methodological guidance and research support in collecting interview and social media data and developing data mining and analysis methods for their infrastructure planning and healthcare research.

- Mentored Emmanuel Edusei on a **coauthored manuscript** currently under review at *Cities*.

Honors and Awards

2026	**GPSS Travel Fund Program Award** ($500) (2%), Virginia Tech	USA
2024	**Myers-Lawson Doctoral Fellowship** (4%), Virginia Tech	USA
2022	**National Scholarship** (1%), Ministry of Education of China	China
2020–2022	**First Prize, Graduate Scholarship**, Chongqing University	China
2019	**Third Prize**, Mathematical Contest in Modeling (MCM/ICM)	USA
2018	**Third Prize, Undergraduate Scholarship**, Chongqing University	China

Service to the Profession

Professional Membership

- Member, ASCE Visualization, Information Modeling, and Simulation (VIMS) Committee
- Member, ASCE Data Sensing and Analysis (DSA) Committee

Conference Service

- Contributor to the organization of the workshop "Empowering Young Researchers in AEC: Building the AI Competencies of Tomorrow," ASCE International Conference on Computing in Civil Engineering (i3CE 2026), Incheon, South Korea, June 14, 2026

Reviewer

- Engineering, Construction and Architectural Management (ECAM), Emerald
- ASCE International Conference on Computing in Civil Engineering (i3CE), 2025–2026
- Construction Institute & Construction Research Congress Joint Conference, 2026
- 23rd International Conference on Smart and Sustainable Built Environment (SASBE), 2026

Professional Experience

Hong Kong Center for Construction Robotics, Hong Kong — Aug 2023 – Dec 2023
Research Assistant, Construction Robotics

- Developed automated perception functions for construction robots, including panel recognition and LiDAR-based obstacle avoidance.
- Prototyped optimization-based route planning for autonomous robot navigation on construction sites.
- Implemented in Python on Linux with the Robot Operating System (ROS).

Skills

AI & Machine Learning	Prompting, multi-agent systems, deep learning
Programming	Python, R, Linux
BIM & 3D Modeling	Revit, IFC, point cloud processing
Web & XR	JavaScript, three.js
Robotics	ROS
Languages	English, Chinese (native language)